Mr Dimelow

3 Processes, techniques and specialist tools

3.5 Application, advantages and disadvantages of the following finishing techniques and methods of preservation in order to discriminate between them and select appropriately for use, including for the prevention of degradation:

a) ***finishes*** – *paints, varnishes, sealants, preservatives, anodising, electro-plating, powder coating, oil coating, galvanisation, cathodic protection (including use of specialist tools)*

Finishes

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Finishes

Whilst bulk material properties will provide many of the characteristics required for your product, you may choose to coat or plate the surface for functional or cosmetic reasons.

Painted Coatings

Use on ***metals / woods***.

Paint *(**cellulose** (for metals) and **acrylic**)* can provide some protection against ***corrosion***, as well as ***cosmetic benefits*** *(colour / textured finish).* Appropriate paints and coatings can be applied to most surfaces and can also be applied in ***layers***, with each different coating adding a different function – for example, *one layer might provide corrosion resistance, another could act as a base for a cosmetic coat, and the final layer could be a lacquer over a colour or metallic coating.*

Paint may be applied using conventional ***brushes*** or ***strayed*** on.

Advantages:

Paints are primarily used on low-cost metals such as steel and low cost wood such as pine.

Disadvantages:

- *Metal must be degreased before paint is applied.*
- *Wood knots should be treated with knotting to prevent the resin spoiling the finish.*
- *Undercoats of colours are often needed for both wood and metal.*



Cellulose Paint





Acrylic Paint



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Painted Coatings



Varnish

Polyurethane or ***Acrylic varnish*** used for ***wood*** finishes.

Features:

- *Available in gloss, satin and matt finish. Available in clear or coloured.*
- *Hard, heatproof, waterproof finish.*
- *Used on furniture and some outdoor products.*

Application:

Apply a thin coat with a brush in the direction of the grain.

Several thin coats can be applied, sanding between coats.

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Advantages:

Provides a hard, tough, heatproof, waterproof finish.

Clear finish allows the grain of the wood to be seen through the finish.

Disadvantages:

Wood may need sanding *(glass papering)* between coats.





Polyurethane Varnish

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Sealants

Used for ***metals***.

Features:

- *Sealant is polymer-based and protects polished surfaces from corrosion.*
- *Used in the automotive industry as an additional bodywork barrier against the effects of weathering, road salts and insect attack. Used on machine beds to provide a smooth surface.*

Application:

Applied with a ***cloth*** or ***machine pad*** to produce a film that is ***cured*** for up to 15 minutes, then buffed with a cloth to shine.

Advantages:

Increases toughness to the surface.

Disadvantages:

Mostly used as an additional barrier finish, e.g. on top of a paint finish.



Sealants





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Preservations

Used for ***metals***.

Features:

- *Can provide medium to long term protection for metal surfaces.*
- *Often used on moulds and dies for preventing fingerprints showing on surfaces, minor atmospheric corrosion or condensation build up.*

Application:

Applied by wiping on with a cloth, spray or immersion of the product.

Advantages:

Easy to apply, little specialist tooling.

Disadvantages:

None listed.

Oil Coating

Used for ***woods***.

Features:

- *Mostly available in clear.*
- *Danish Oil or Scandinavian Oils are most common.*
- *Teak oil is used for naturally oily woods such as teak.*

Application:

- Apply with a lint-free cloth, rub the oil into the surface of the wood in the direction of the grain.
- Let the wood absorb the oil for five minutes then rub away the excess oil with a clean cloth.
- Lightly sand (glass paper) or rub with wire wool to open up the grain between coats.

Advantages:

Easy process enhances the grain, improves water, decay, insect and fungal resistance.

Disadvantages:

- *May require minimal sanding (glass papering) between coats.*
- *Needs reapplying periodically.*
- *Not very hardwearing.*









Oil Coatings

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Anodising

Used for ***Aluminium*** *(only).*

Features:

- *Improves hardness, can also provide colour.*
- *Used in the outer casings of torches, external hard drives covers and casings for mobile phones or digital music players.*

Application:

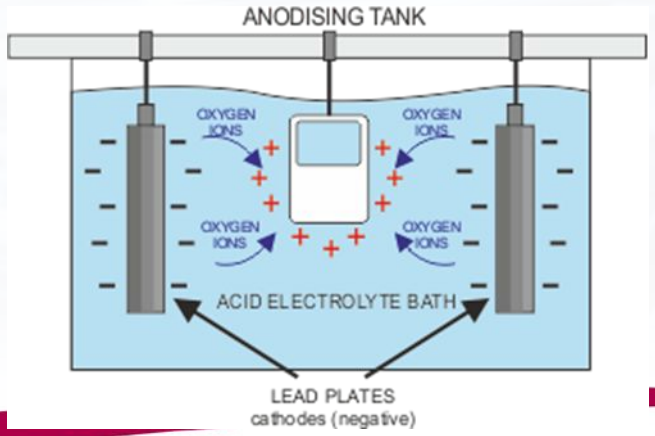
- *Similar setup to electroplating.*
- *Electric current passes through a sulphuric acid electrolyte solution from the part to be treated to a negative cathode.*
- *Current flows from the part to be treated (positive anode) to the negative cathode, the aluminium oxide layer builds up on the treated part, producing the anodised finish.*

Advantages:

- Enhances the natural oxide layer, giving increased hardness and toughness.
- Anodised products can be finished with clear lacquer to seal the finish and provide further protection against surface scratches.

Disadvantages:

- Requires a sulphuric acid electrolyte solution bath for the process and electric current.
- Specific process for aluminium.



Anodised Components



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Anodising Video



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Electroplating

Used on ***metals***.

Features:

Electroplating is a an electrochemical process by which a ***base material*** is given a ***surface coating*** designed to improve *corrosion resistance, wear or cosmetic characteristics*.

Probably the best-known electroplating process is ***chrome plating***, which can give both decorative and wear-resistant surfaces. Other materials used for electroplating include *silver, gold, zinc, copper and tin.*

Electroplating *is a process where a coating of metal is added to a* ***conductor*** *using* ***electricity***.

Electroplating *is also known simply as* ***'plating'***. *When a current is applied to the conductor to be coated,* ***metal ions*** *in solution are reduced onto the electrode to form a thin layer.*

Electroplating

Used on ***metals***.

Application:

- *Product and donor material placed in a container with electrolyte solution.*
- *Direct current (DC) is applied, the product attracts the donor metal and the product is electroplated.*

Advantages:

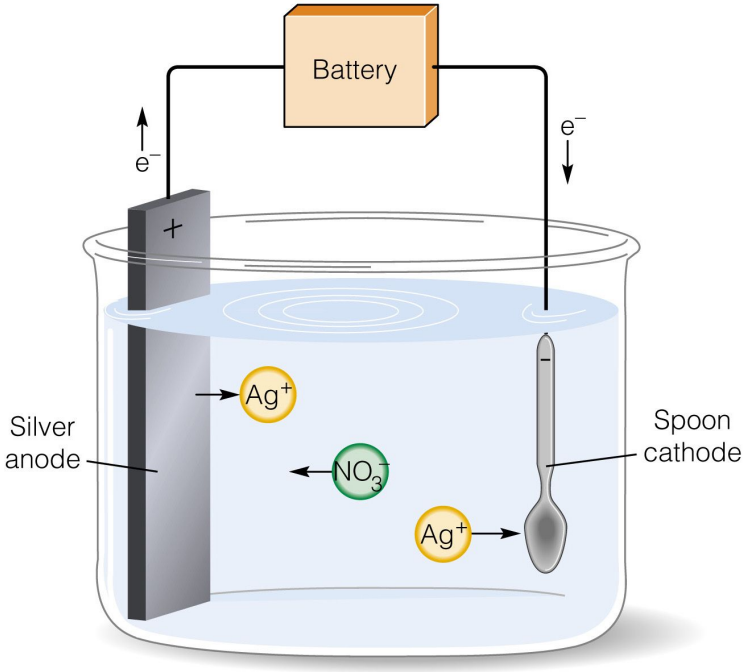
- *Used on low cost metals.*
- *Makes surface more hardwearing.*
- *Provides colour.*

Disadvantages:

Needs container with electrolyte solution and electric current.

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Electroplating









Electroplated Components





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Electroplating Video



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Galvanising

Used on ***metals***.

Features:

Galvanising consists of ***zinc plating*** and is a form of coating applied to ***iron*** or ***steels parts***. It often gives a characteristic crystalline ***'spangled'*** surface finish, which can be seen on galvanised steel buckets, for example.

Galvanising offers ***sacrificial protection*** in that the ***zinc*** will corrode before the underlying steel, and it will still provide protection even where the coating has been scratched through to the steel.

Application:

Metal is coated by dipping the product into a tank of ***molten zinc*** at approximately 460°C.

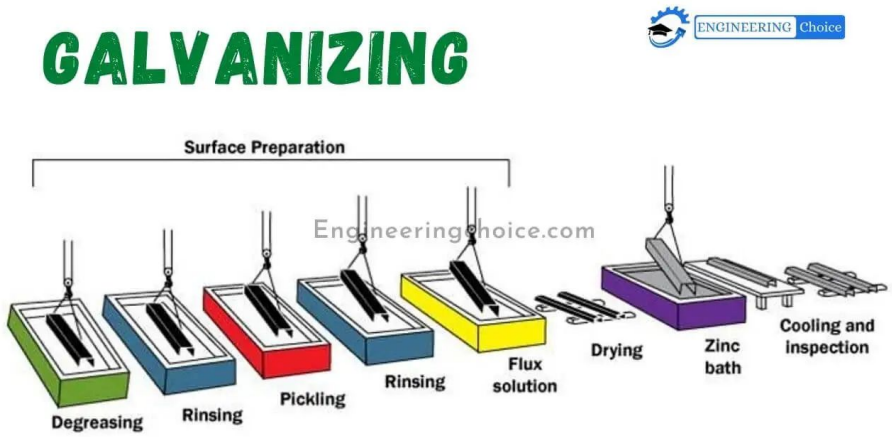
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Advantages:

- *Provides a barrier against corrosion.*

Disadvantages:

- *Prior to the dipping process, the metals must be cleaned and degreased.*
- *Need a dipping bath.*





Galvanised Components



Galvanising Video



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Powder Coating

Used on ***metals***.

Features:

Powder coating offers many of the same benefits as conventional paint but can result in a ***tougher surface***.

Used on *washing machine, dishwasher and refrigerator bodies (cases), also for metal gates, fencing and railings*.

Application:

- The ***product*** to be coated is ***negatively charged***.
- ***Polymer powder (positively charged)*** is then sprayed through an airgun and it is attracted to the product ***(electrostatic charge)***.
- The ***powder*** is then ***cured (baked)*** by placing the parts in an ***oven*** to yield a hard even surface. The heat melts the powder over the product to give an even coating.

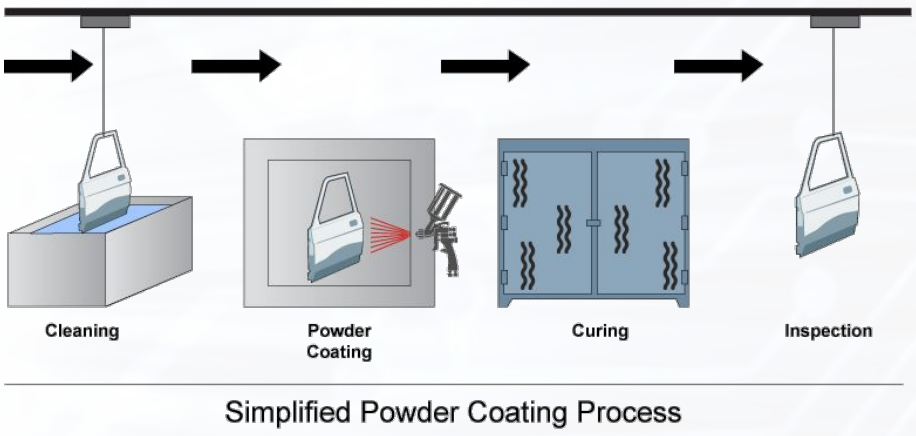
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Advantages:

- *More hardwearing than paint.*
- *Provides colour.*
- *Good for low cost metal such as mild steel.*

Disadvantages:

- *Relatively simple process but needs electrical charge and oven of appropriate size for the product.*



Simplified Powder Coating Process

Powder Coated Components



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Powder Coating Video



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Dip Coating

Dip coating is a process by which ***base material*** is submerged in ***polymer coating*** *(e.g. PVC, nylon or polyethylene)*, then taken out and allowed to drip dry. Once the excess coating has drained, the substrate material is further dried using one of a variety of techniques, often baking.

Advantages / Disadvantages of Dip Coating

While dip coating is a simple, reliable, and powerful option, capable of coating almost any substrate material, it also has its weaknesses.

These include unbalanced coverage (varying thickness across the surface of the material), or coating build-up (which is created as the material "drip-dries").

Dip Coating

There are ***three*** *essential parts* in the ***dip coating process***: immersion, dip time, and removal.

1. ***Immersion*** *is the first step and is exactly what it sounds like: immersing the substrate material in the conformal coating. During this stage, it's important to maintain steady speeds and smooth, unshaken conditions.*

2. ***Dip time*** *refers to the time the substrate material remains immersed in the coating. Limiting movement is especially important during this step, or coating could become uneven and unreliable.*

3. ***Removal*** *is the final step, and, in many ways, the most important one: faster removal will result in a thicker coating, so it's important to time things carefully.*

Dip Coating



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Cathodic Protection

Used on ***metals***.

Features:

Protects steel structures buried in soil or immersed in water, such as pipelines, ship hulls, jetties and offshore platforms.

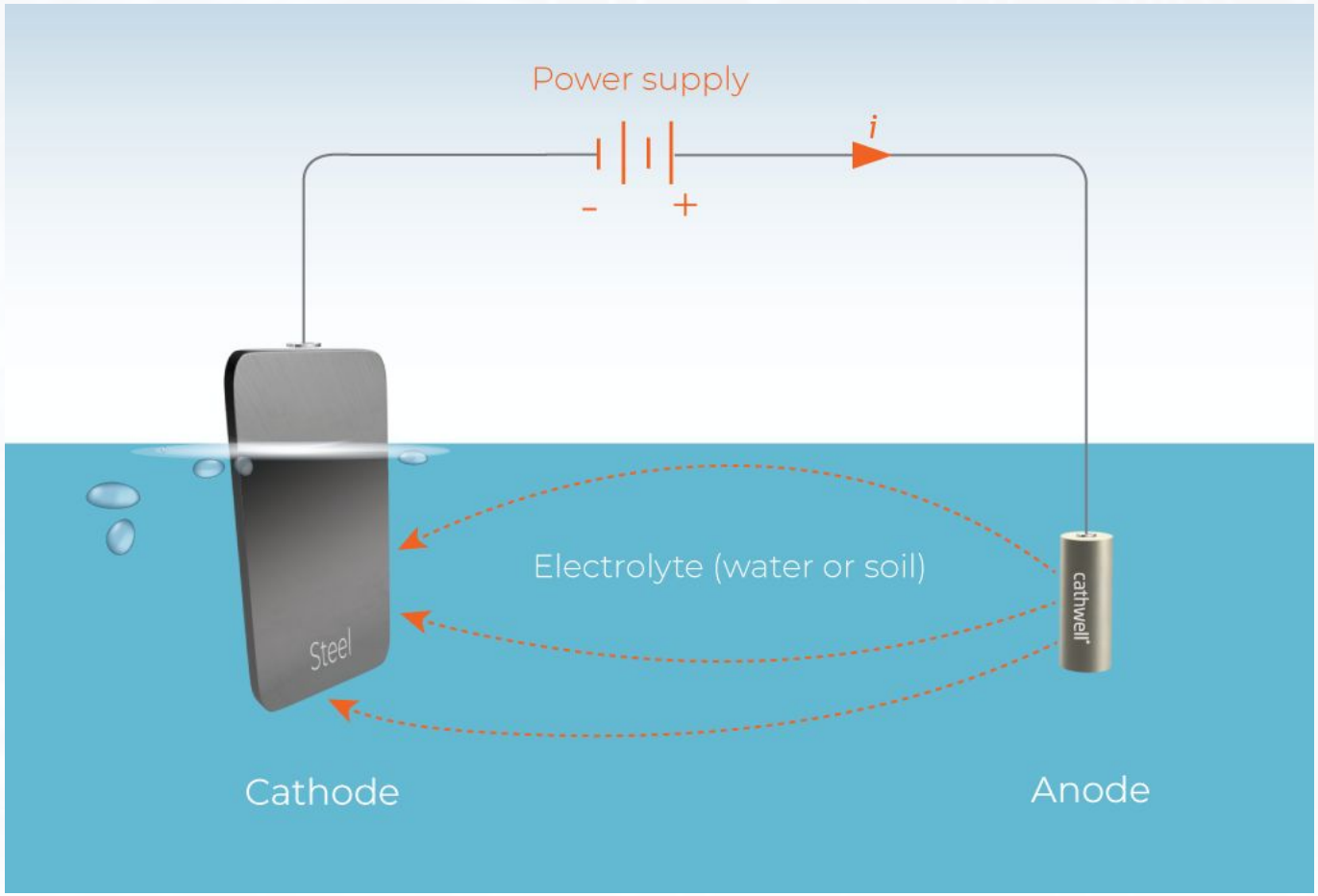
Application:

Impressed current:

- The part to be protected is connected to an ***electrical power supply***. The current flows from an ***inert electrode*** through any liquid to the component to be protected.
- For example, a buried pipeline receives direct current from an electrode buried in the ground. The pipe becomes the anode and the electrode is the cathode. The cathode corrodes over time and such corrosion is monitored so that the pipeline remains in tact.

Cathodic Protection

Impressed current:



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Cathodic Protection

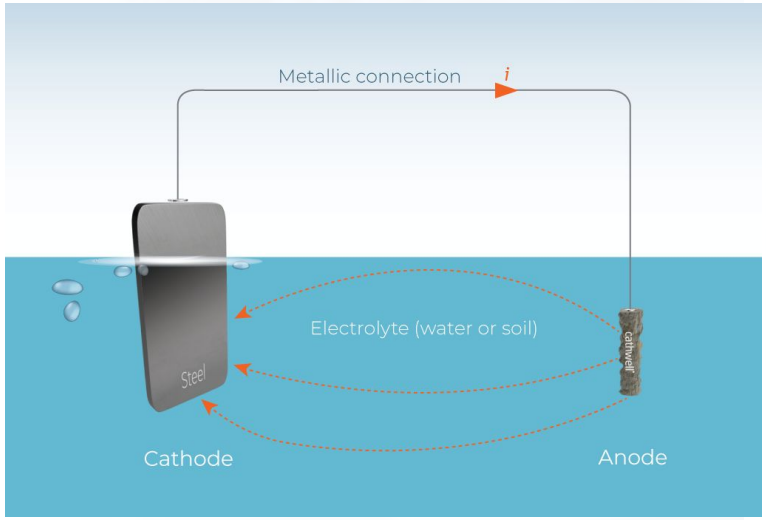
Application:

Sacrificial Anodes:

- This is a more basic method because it uses a sacrificial metal to protect the metal product.
- An electrochemically active metal is joined to the less active metal to provide resistance to corrosion.
- On large structures, the sacrificial metal is monitored for signs of corrosion and replaced when the base metals is no longer protected.

Cathodic Protection

Sacrificial Anodes





The sacrificial anodes corrode themselves protecting the Hull. These dissolving anodes create a corrosion current in the system due to a potential difference created between the anode and hull material.

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Cathodic Protection

Application:

- ***Sacrificial Anodes:*** this is a more basic method because it uses a sacrificial metal to protect the metal product.

- An electrochemically active metal is monitored for signs of corrosion and replaced when the base metals is no longer protected.

Advantages:

- *Good for underground or underwater products, large scale products.*

Disadvantages:

- *Products may need frequent monitoring for Signs of corrosion.*

Typical Mistakes

Forgetting to explain how the finish is applied or what material it is used for.

Try to structure your answer to state the finish, the material it is used on and then discuss any features, advantages etc…

Exam Tip

Know the specific finishes, their advantages and what material they are used to protect.

Task1: Surface Treatments and Coating

Research the ***'surface treatments'*** listed below and produce a report. Illustrate your answers with diagrams or images were possible.

Your report should give examples of the *materials* and *products* which are *'protected'* by these *'surface treatments'*, *how they are applied and the advantages and disadvantages of each*.

1. ***Painted Coatings***
2. ***Varnishing***
3. ***Oil Coating***
4. ***Powder Coating / Dip Coating***
5. ***Anodising***
6. ***Electroplating***
7. ***Galvanising***
8. ***Catholic Protection***

Task2: Test Yourself

1. State ***two*** methods used to apply paint to a product.
2. Describe the process of ***powder coating***.
3. State ***four*** metals that can be coated onto steel by ***electroplating***.
4. Name ***three*** steel products that are ***galvanised***.
5. Give ***two*** reasons for ***anodising*** an aluminium product.

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Task2: Test Yourself

6. Name the *material* used in ***anodising*** and a ***product*** that has an ***anodised finish***.
7. Suggest a *suitable finish* for *washing machine body panels*.
8. What ***sacrificial metal*** is used in the ***galvanising*** process?

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